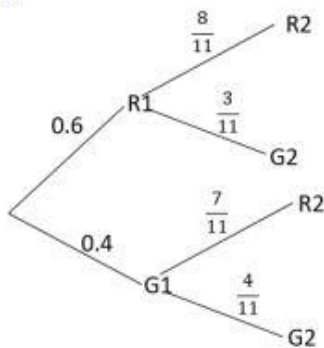


1.



a. $P(R2) = (0.6)\left(\frac{8}{11}\right) + (0.4)\left(\frac{7}{11}\right) = 0.69$

b. $P(R1|G2) = \frac{P(R1 \cap G2)}{P(G2)} = \frac{(0.6)\left(\frac{3}{11}\right)}{(0.6)\left(\frac{3}{11}\right) + (0.4)\left(\frac{4}{11}\right)} = 0.53$

2.

a. $P(V \cap W) = P(V) + P(W) - P(V \cup W) = 0.15 + 0.05 - 0.17 = 0.03$

b. $P(V^c \cap W^c) = 1 - P(V \cup W) = 1 - 0.17 = 0.83$

c. $P(V \cap W^c) = P(V) - P(V \cap W) = 0.15 - 0.03 = 0.12$

3. Categorical – gender, vote Quantitative – age, household income

4. $P(\text{both fail}) = P(F \cap F) = (0.22)(0.22) = 0.0484$

5.

a. Outliers: none

Center: $\bar{x}=766.6$ $M=784$

Spread: $s=91.5$ $IQR=132$

#peaks: Unimodal or Bimodal

Symmetry: left skewed

b. Reduces rainfall, mean goes down substantially (766.6 vs 850).